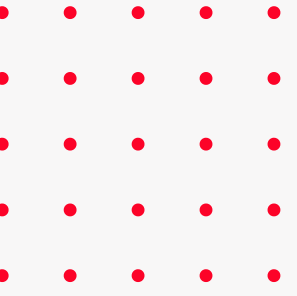


Customer Behaviour Analysis – Retail & E-commerce



Contents



01 Project Introduction & Context

02 Business Problem & Data Preparation

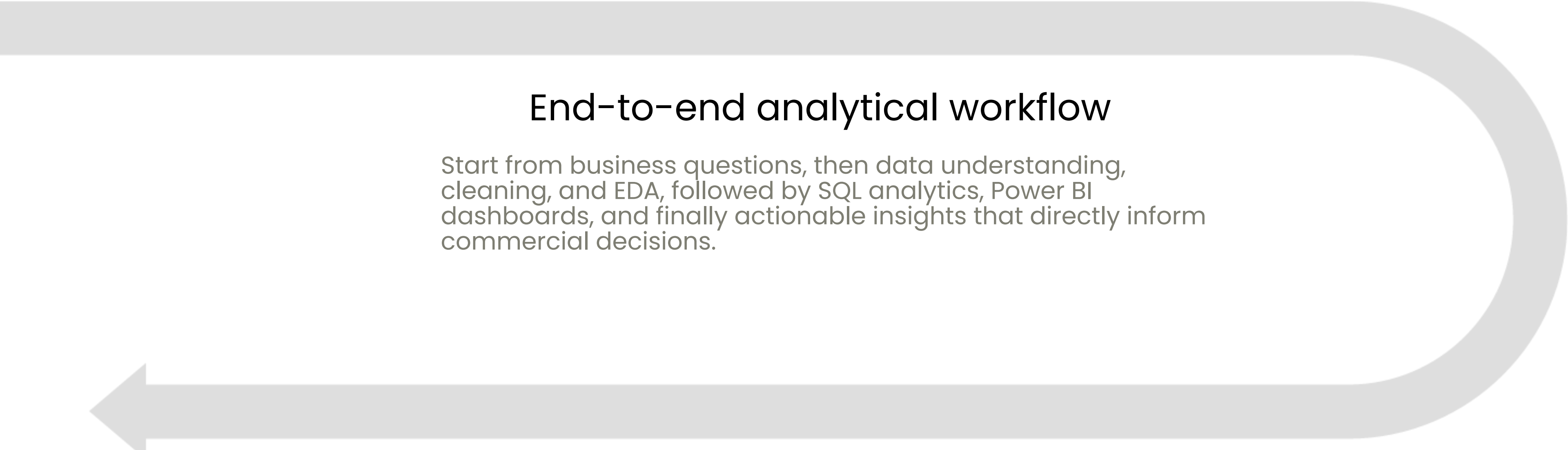
03 Analysis & Insights

04 Dashboards, Decisions

01 Project Introduction & Context



Analytical Approach & Tools



End-to-end analytical workflow

Start from business questions, then data understanding, cleaning, and EDA, followed by SQL analytics, Power BI dashboards, and finally actionable insights that directly inform commercial decisions.

Tool stack and reproducible practices

Python handles preparation and EDA, SQL powers aggregation and querying, Power BI delivers storytelling dashboards, supported by clean modular code, Git-based version control, and portfolio-grade documentation.

Dataset Overview

Structure & Key Fields

Transaction-level retail and e-commerce dataset where each row represents one purchase, keyed by Customer ID, Transaction ID, and Date, enabling detailed temporal, customer-level behavior tracking.

Variables, Volume & Source

Includes numeric fields (Age, Purchase Amount, Discount %, Previous Purchases, Basket Size) and categorical dimensions across product, payment, season, subscription, gender, location; anonymized, simulated portfolio dataset covering 24 months.

Field Name	Type	Description
Customer ID	Categorical	Unique identifier for each customer
Item_Purcahsed	Categorical	Sub_category itme purchased
Category	Date	Category
Age	Numeric	Customer age at time of transaction
Purchase Amount	Numeric	Total monetary value of the transaction
Discount Applied	Numeric	Percentage discount applied to the transaction
Previous Purchases	Numeric	Count of prior transactions by the same customer
Size	Numeric	Number of items included in the transaction
Review ratings	Numeric	Review on item purahsed
Payment Method	Categorical	Payment Methods used for shopping
Season	Categorical	Seasonal label (e.g., Spring, Summer, etc.)
Subscription Status	Categorical	Subscripitiion customer or not
Gender	Categorical	Self-identified gender of the customer
Location	Categorical	Customer's geographic region or store location
Frequency of Purchases	Categorical	how often customer visit

02 Business Problem & Data Preparation



Business Problems & Objectives

Low Customer Retention

Who are frequent buyers vs. one-time buyers?

Product Demand Understading

Which category performs best? Which products generate the highest revenue?

Seasonal Demand Variation

Does season impact sales? Which season drives maximum purchases?

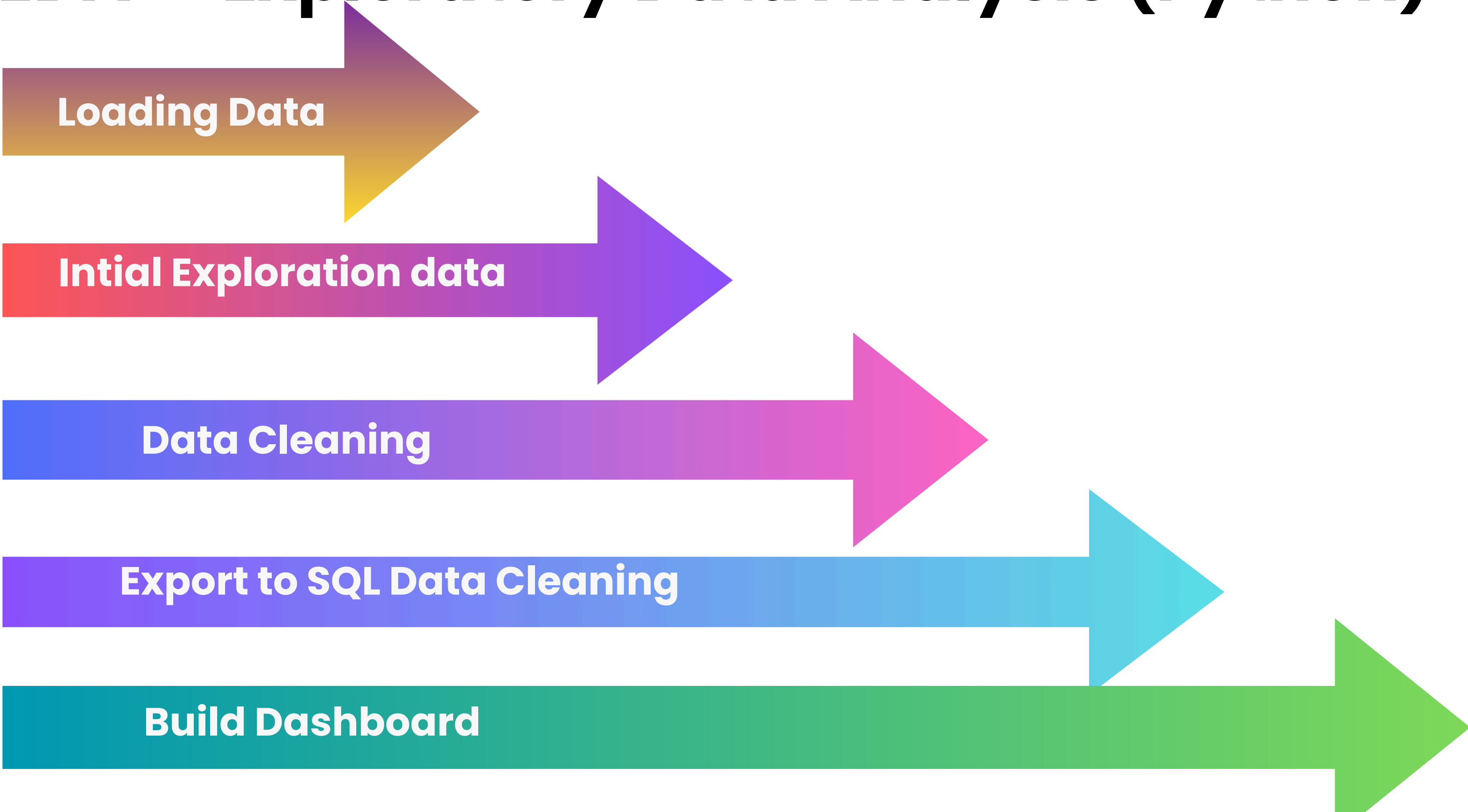
Ineffective Marketing Campaigns

Do discounts increase purchases? Which customers respond to promo codes?

Customer Segmentation

Identify high-value, frequent, and discount-sensitive customers

EDA — Exploratory Data Analysis (Python)



Loading Data

Initial Exploration data

Data Cleaning

Export to SQL Data Cleaning

Build Dashboard

Why This Data Matters



High-Value Customers

Identify frequent buyers and one-time purchasers



Purchase Patterns

Understand what, how often, and how much customers buy



Discount Strategy

Optimize promo codes and campaign effectiveness



Customer Lifetime Value

Identify frequent buyers and one-time purchasers

03 Analysis & Insights



Customer Segmentation – New Customers or Repeat once

Insights, Customer Category Impact
& Actions

Which customer is most valuable and CLV for business growth and types of customers perform in revenue.

-- ? Business Problem :-
Unclear relationship
between loyalty and subscription.



Impact:-

- Improves subscription targeting
- Increases conversion to paid programs
- increase customer retention strategy

```
select count(*) as customer_count,  
subscription_status,  
case when previous_purchases > 5 then "Repeat Buyers"  
else "Normal Buyers"  
end as Customer_type  
from customer_behaviour  
group by 2,3;
```

	customer_count	subscription_status	Customer_type
▶	2763	No	Repeat Buyers
	1190	Yes	Repeat Buyers
	658	No	Normal Buyers
	389	Yes	Normal Buyers

Customer Segmentation – Loyal, Returning, New

Insights, Customer Category Impact & Actions

Which customer is most valuable and CLV for business growth and types of customers perform in revenue.

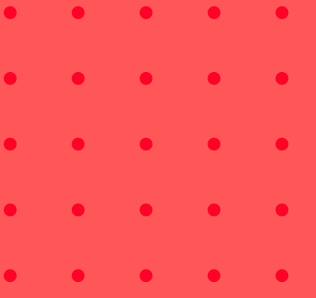
? Business Problem :-
Lack of customer segmentation leads to generic strategies.

-  Impact:-
- Enables personalized marketing
 - Improves retention strategies
 - Increases conversion rates

```
select
case
when previous_purchases = '0' then "New Customer"
when previous_purchases between 1 and 15 then "Returning customer"
else "Loyal Customer"
end as customer_segment,
count(*) as "Total_Customers"
from customer_behviour
group by customer_segment ;
```

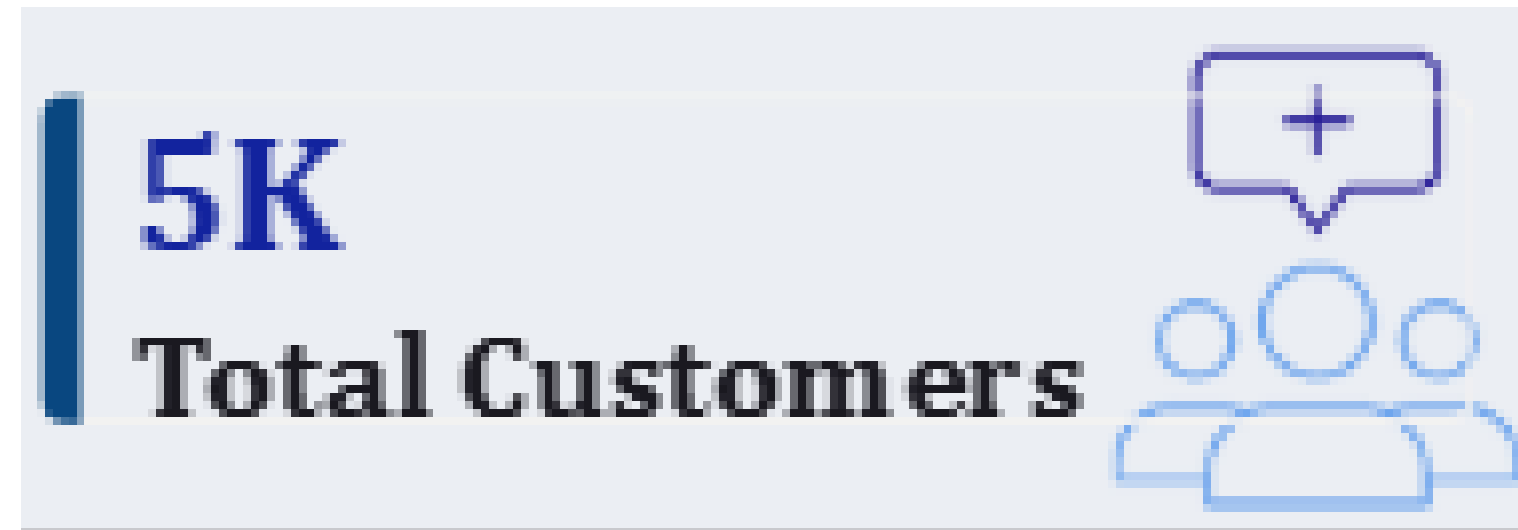
	customer_segment	Total_Customers
▶	Loyal Customer	3085
	Returning customer	1354
	New Customer	561

04 Dashboards, Decisions



Executive KPIs

KPI stands for Key Performance Indicator — a measurable value used to track how effectively a person, team, or organization is achieving a goal.



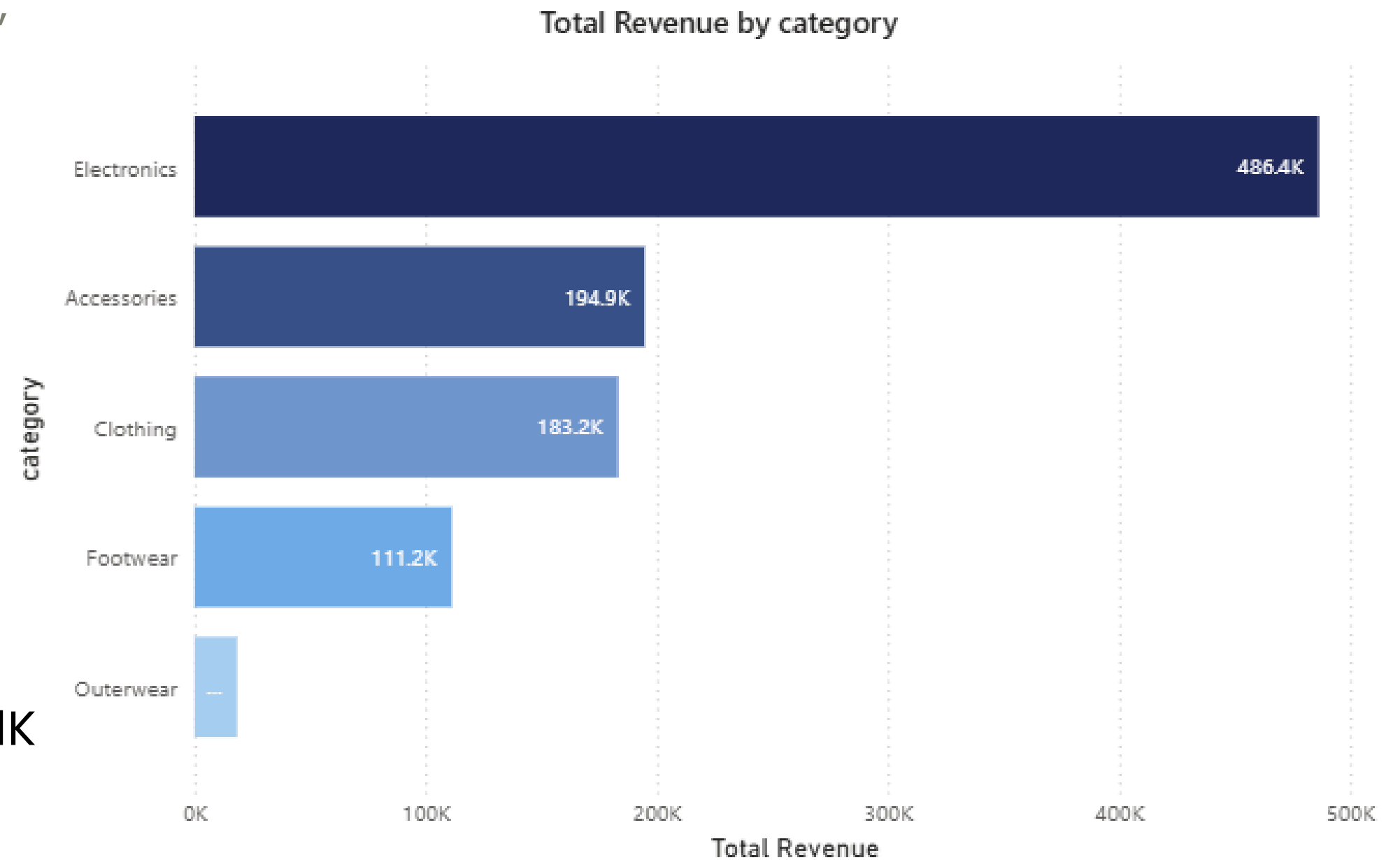
Power Bi Revenue & Category Performance

01 Category Revenue Overview

Bar chart highlights revenue distribution across Electronics, Clothing, Accessories, Footwear, Others. Electronics leads overall revenue; Clothing and Footwear contribute strong volumes, supporting diversified category performance.

Key Findings

- ✦ Electronics leads revenue at \$486K
- ✦ Discounts increase avg.purchse :\$219 (discount) vs \$182 (no discount)
- ✦ Male: \$391K · Female: \$352K · Other: \$251K revenue
- ✦ Outwear leads revnenu less



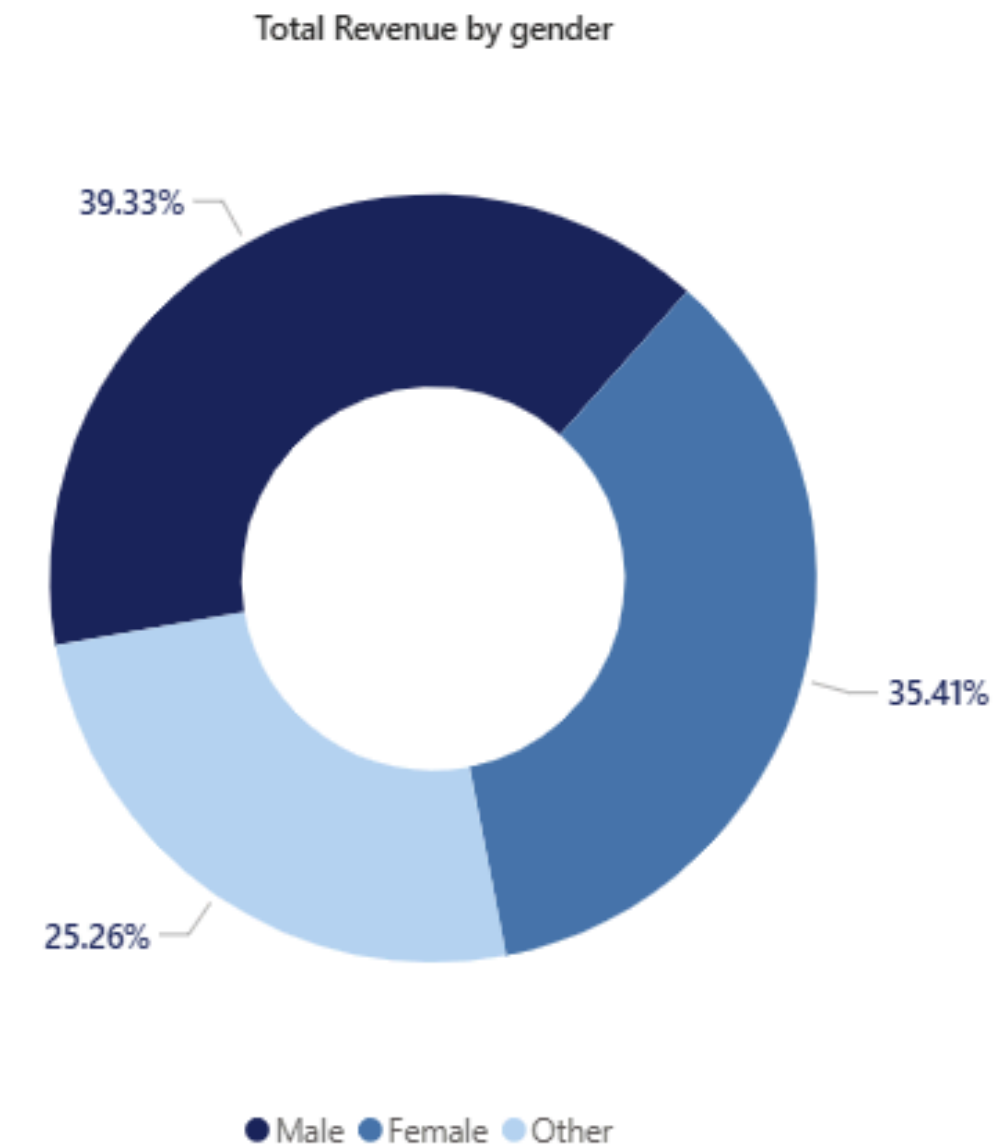
PowerBI Gender Performance

01 Category Revenue Overview

Donut Chart to show the contributions of gender in total revenue by each gender in percentage

Key Findings

- ✦ 39.33% Male is performing to contribute in total revenue
- ✦ 25.26% Other gender



Operational Performance

01 Shipping Revenue Overview

Bar chart highlights Purchase amount and total customer distribution across Different types of shipping type like : Standard, Express, 2-Day shipping, Next Day



Key Findings

- ◆ Standard and Express are almost same performer 1201 Total Customer in Standard shipping 1199 in Express
- ◆ 2-Day Shipping operations has lowest number of customer 627 Total Customers Only
- ◆ Standard and Express are contributing to most revenue

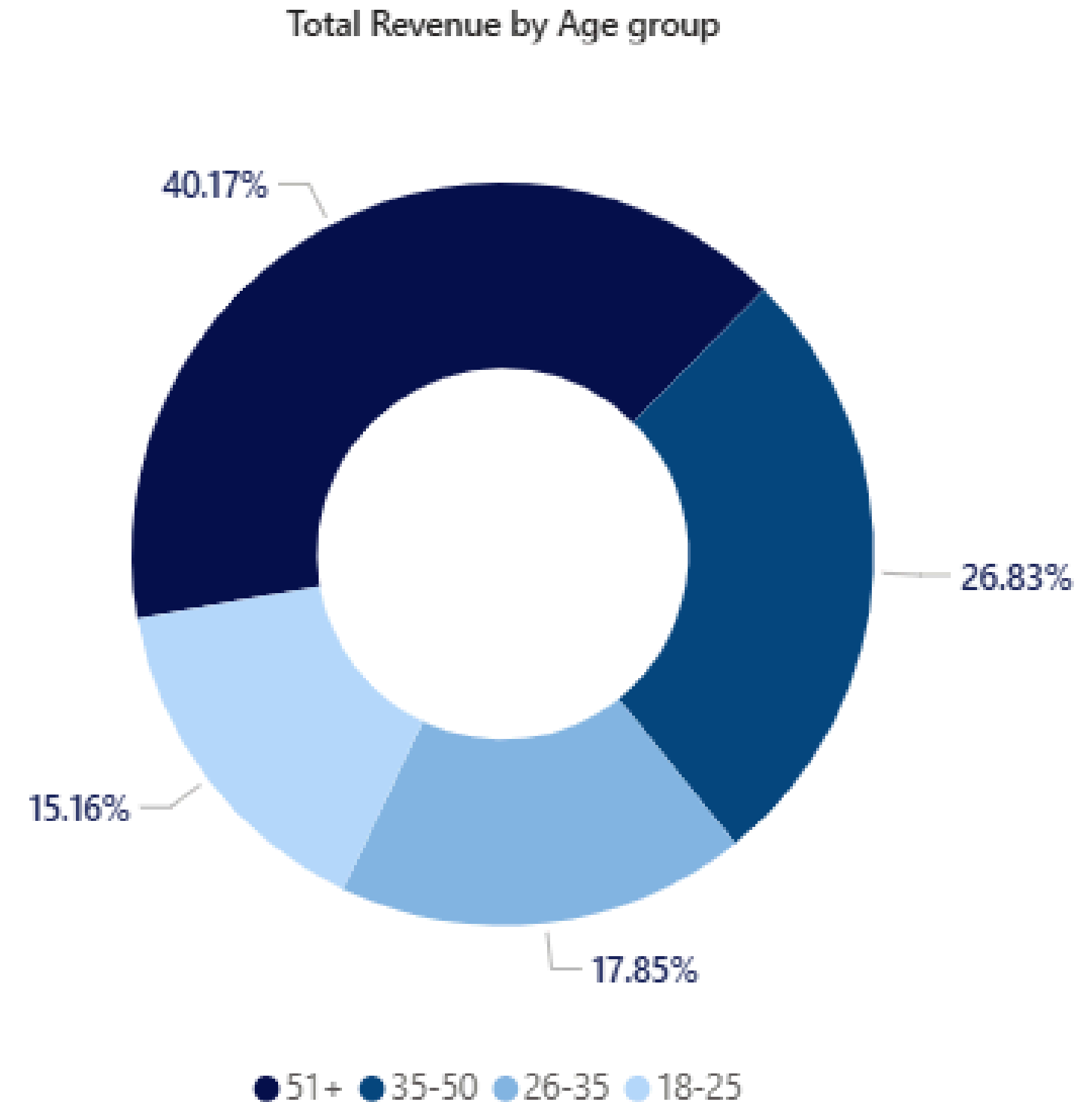
Operational Performance

01 Shipping Revenue Overview

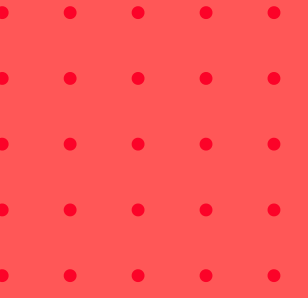
Donut chart highlights Total revenue amount Percentage across Different age group

Key Findings

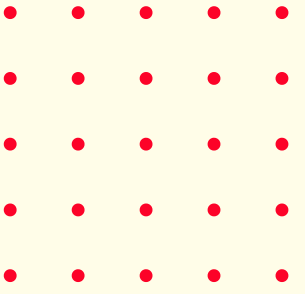
- ✦ Age 51+ Highest customer which contribute to Total revenue 40.17%
- ✦ The Customers Age Between 18-25 Lowest customer which contribute to Total revenue 15.16%



05 Conclusions



Final Business Conclusion (Project Conclusion)



Customer Behavior Analysis – Final Business Conclusion

The analysis reveals that customer purchasing behavior is strongly influenced by demographic factors, product preferences, and spending patterns. A small group of high-value customers contributes a significant share of total revenue, indicating the importance of customer retention and personalized marketing strategies.

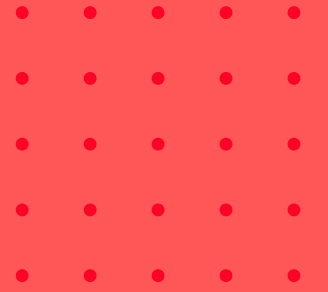
Revenue analysis shows that certain customer segments, age groups, and product categories consistently generate higher sales, while low-performing segments contribute less to business growth. This suggests that the company should focus more on profitable customer groups and optimize marketing campaigns for underperforming segments.

Business Recommendations

- 1.Focus retention strategies on high-value and repeat customers.
- 2.Increase personalized marketing using customer segmentation.
- 3.Promote best-selling products more aggressively.
- 4.Reduce dependency on unnecessary discounts.
- 5.Improve engagement for low-performing customer groups.
- 6.Use customer purchase behavior to forecast future sales and inventory demand.

Final Outcome

Overall, the dashboard helps management make data-driven decisions by identifying revenue drivers, customer trends, profitable segments, and sales opportunities, ultimately supporting higher profitability, improved customer retention, and smarter business growth strategies.



Thanks

Gratitude, Appreciation, and
the Power of Thanks